

POC Test Cases — Data Platform Validation



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Info

This document demonstrates the `testbook.cls` template for Huawei Cloud POC/acceptance test case documents. It exercises all available commands and environments: cover page, table of contents, test case environment with all field commands, objectives block, code blocks, images, callouts (`warning`, `tip`, `infobox`), tables, badge, menu paths, weblinks, notes, param references, test summary table, test steps tables, result badges, and changelog.

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Test Cases Description

General Objective: Validate the Huawei Cloud Data Platform deployment for the POC environment, ensuring that all core services (DataArts Studio, MRS, OBS, VPC) are provisioned correctly, accessible, and functioning as specified in the solution architecture.

Prerequisites:

- Huawei Cloud account with IAM admin privileges in the `sa-brazil-1` region.
 - VPC, subnet, and security groups already created by the infrastructure team.
 - DataArts Studio instance (Dayu) provisioned and licensed.
 - MRS cluster deployed with the required component stack (HDFS, YARN, Spark, Hive).
 - OBS bucket created and accessible with the designated IAM policy.
 - Test execution environment with network connectivity to the Huawei Cloud Console.
 - `latexmk` and `terraform` CLI tools installed locally.
-

1.1 Test Scope

Domain	What Should Be Observed
Platform Architecture	Core services are deployed, healthy, and reachable via the Console API.
Data Engineering	DataArts Studio pipelines can ingest, transform, and load data into the target storage.
Data Storage	OBS buckets and MRS HDFS accept writes/reads with correct permissions.
Security	IAM policies, VPC isolation, and security group rules enforce the intended access control.

Table 1: Test scope by domain.

1.2 Preconditions and Preparations

- All infrastructure resources must be provisioned via Terraform and the `terraform apply` output must show zero errors.
- The MRS cluster must be in **Running** state before any data engineering test case is executed.
- Test data files (CSV/JSON) must be uploaded to the designated OBS bucket prior to pipeline execution.
- Each tester must have a personal IAM account with the **Dayu_User** role assigned in DataArts Studio.
- Network firewall rules must allow inbound TCP 22 (SSH) and TCP 443 (HTTPS) from the test execution subnet.



Ensure the test execution environment has the correct timezone set to America/Sao_Paulo (GMT-3) so that log timestamps align with the Huawei Cloud Console.

```
# Verify CLI tools are installed
terraform version
obsutil version
kubectl version --client
```

Tip

Use `terraform plan` before `terraform apply` to preview infrastructure changes and avoid unintended resource modifications.

1.3 Acceptance Method

Result	Explanation
Satisfied	The test case passes fully — observed behavior matches the expected result with no deviations.
Satisfied but with reservations	The test case passes but minor deviations were observed that do not impact the POC objectives (documented in Remarks).
Not satisfied	The test case fails — observed behavior does not match the expected result. A defect report must be filed.
Untested	The test case could not be executed due to blocking dependencies or environment issues.

Table 2: Acceptance criteria.

1.4 Test Case Summary

ID	Title	Status
TC-001	Verify MRS cluster deployment and health	Pass
TC-002	Verify OBS bucket creation and accessibility	Pass
TC-003	Verify VPC and security group configuration	Pass
TC-004	Verify DataArts Studio workspace creation	Pass
TC-005	Verify data ingestion pipeline execution	Pass
TC-006	Verify Spark SQL query on MRS cluster	Pass
TC-007	Verify IAM policy enforcement on OBS buckets	Pass
TC-008	Verify DataArts Studio data masking rule	Pass
TC-009	Verify Terraform infrastructure deployment	Pass
TC-010	Verify test data upload to OBS	Pass



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Platform Architecture



2.1 TC-001: Verify MRS cluster deployment and health

Objective	Confirm that the MapReduce Service (MRS) cluster is deployed with the required components (HDFS, YARN, Spark, Hive) and all nodes report a healthy status.		
Prerequisites	1. Terraform infrastructure apply completed successfully. 2. MRS cluster ID is available in the Terraform output. 3. IAM user has MRS_Viewer or higher role.		
Procedure	See test steps below.		
	1. Cluster status is Running .		
Step	Action	Expected Result	Status
1	Log in to the Huawei Cloud Console	Console dashboard displayed	Pass
2	Navigate to Console → MapReduce Service → Clusters	Clusters list page shown	Pass
3	Locate cluster <code>mrs-poc-cluster</code>	Cluster found in list	Pass
4	Verify cluster status	Status is Running	Pass
5	Review Component tab	HDFS, YARN, Spark, Hive all Normal	Pass
6	Review Node tab	All nodes show Running	Pass
Expected Result	 2. All four components (HDFS, YARN, Spark, Hive) are listed with status Normal. 3. All nodes report status Running with no alarms.		
Remarks	If any component shows Abnormal , check the MRS alarm page before marking the result.		

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2.2 TC-002: Verify OBS bucket creation and accessibility

Objective	Confirm that the Object Storage Service (OBS) bucket used for the POC data lake is created, has the correct storage class, and is accessible with the designated IAM policy.
Prerequisites	1. OBS bucket name is defined in the Terraform configuration.
2. IAM policy granting read/write access to the bucket is attached to the test user.	
Procedure	1. Log in to the Console.
2. Navigate to Console → Object Storage Service.	
3. Locate the bucket (e.g. poc-datalake-raw).	
4. Verify the storage class is Standard.	
5. Upload a test file (test-upload.txt) to the bucket.	
6. Download the file and compare its content with the original.	
7. Delete the test file.	
Expected Result	1. Bucket exists with storage class Standard .
2. File upload completes without error.	
3. Downloaded	



2.3 TC-003: Verify VPC and security group configuration

Objective	Confirm that the VPC, subnet, and security groups are configured according to the network architecture diagram, with the correct CIDR blocks and inbound/outbound rules.
Prerequisites	1. VPC and subnet IDs are available from the Terraform output.
2. Security group names are documented in the solution architecture.	
Procedure	1. Log in to the Console.
2. Navigate to Console → Virtual Private Cloud → VPCs.	
3. Locate the VPC (e.g. vpc-poc) and verify its CIDR block matches the architecture (e.g. 10.0.0.0/16).	
4. Click the subnet and verify its CIDR (e.g. 10.0.1.0/24).	
5. Navigate to Security Groups and verify inbound rules: TCP 22 from the bastion subnet, TCP 443 from the corporate proxy.	
6. Verify outbound rules allow all traffic (default).	
Expected Result	1. VPC CIDR is 10.0.0.0/16.
2. Subnet CIDR is	



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Data Engineering



3.1 TC-004: Verify DataArts Studio workspace creation

Required Objective	Confirm that the DataArts Studio (Dayu) instance is provisioned, the workspace is accessible, and the designated IAM users can log in with the Dayu_User role.
Prerequisites	1. DataArts Studio instance is provisioned and its status is Running .
2. Test IAM user has the Dayu_User role assigned.	
Procedure	1. Log in to the Console with the test IAM user.
2. Navigate to Console → DataArts Studio → Workspaces.	
3. Locate the workspace (e.g. poc-workspace).	
4. Click Access Workspace to open the DataArts Studio console.	
5. Verify the left navigation panel loads with the expected modules: Data Integration, Data Development, Data Architecture.	
Expected Result	1. Workspace is listed and accessible.
2. DataArts Studio console opens without errors.	
3. All three modules (Data Integration, Data	



3.2 TC-005: Verify data ingestion pipeline execution

Objective	Confirm that a DataArts Studio batch pipeline can successfully ingest CSV data from OBS, apply a basic transformation, and write the result to the target OBS path.		
Prerequisites	1. TC-004 completed (DataArts Studio workspace accessible). 2. Source CSV file (<code>customers.csv</code>) exists in the OBS raw bucket. 3. Target OBS path (<code>poc-datalake-curated/customers/</code>) is configured. 4. Pipeline <code>pl-ingest-customers</code> is published in DataArts Studio.		
Procedure	See test steps below.		
	1. Pipeline execution completes with status Success .		
Step	Action	Expected Result	Status
1	Navigate to Data Development in DataArts Studio	Data Development page loads	Pass
2	Locate pipeline <code>pl-ingest-customers</code>	Pipeline found in job list	Pass
3	Click Run to execute the pipeline	Pipeline starts execution	Pass
4	Wait for pipeline status (timeout: 10 min)	Status changes to Success	Pass
5	Verify output file in OBS	File exists in target path	Pass
6	Download and verify row count	Row count matches source	Pass
# Verify the output file in OBS via CLI <pre>obsutil ls obs://poc-datalake-curated/customers/</pre>			



3.3 TC-006: Verify Spark SQL query on MRS cluster

Objective	Confirm that a Spark SQL query can be executed on the MRS cluster to read data from a Hive table and return correct results, validating the integration between MRS and the data lake.
Prerequisites	1. TC-001 completed (MRS cluster healthy). 2. Hive table poc_db.customers exists and contains data loaded by the ingestion pipeline. 3. SSH access to the MRS master node is configured.
Procedure	1. SSH into the MRS master node. 2. Run the Spark SQL CLI: <pre>spark-sql --master yarn --conf spark.sql.hive.convertMetastoreOrc=true -e "SELECT COUNT(*) FROM poc_db.customers;"</pre> 3. Verify the returned count matches the expected row count (10,000). 4. Run a sample query to verify data integrity: <pre>spark-sql --master yarn -e "SELECT customer_id, name FROM poc_db.customers LIMIT 5;"</pre> 5. Confirm the



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Security and Access Control



4.1 TC-007: Verify IAM policy enforcement on OBS buckets

New Objective	Confirm that IAM policies correctly restrict OBS bucket access — authorized users can read/write while unauthorized users are denied, and the bucket policy matches the principle of least privilege.
Prerequisites 2. Two test IAM users: one with OBS_ReadWrite role, one with no OBS permissions. 3. obsutil CLI configured for both users.	1. OBS bucket poc-dataLake-raw exists with the designated IAM policy.
Procedure 2. Using the authorized user, download and verify the file content. 3. Using the unauthorized user, attempt to upload a file — verify the request is denied (HTTP 403). 4. Using the unauthorized user, attempt to list objects in the bucket — verify the request is denied. 5. Review the bucket policy in the Console at Console → Object Storage Service → Bucket Policies and confirm it grants only the intended	1. Using the authorized user, upload a test file to poc-dataLake-raw.



4.2 TC-008: Verify DataArts Studio data masking rule

Objective Prerequisites 2. Data masking rule configured for column <code>customers.ssn</code> (Social Security Number). 3. Test user has <code>Dayu_User</code> role (not admin).	Confirm that sensitive data columns (e.g. PII fields) are masked according to the DataArts Studio data masking rule when queried by non-admin users. 1. TC-004 completed (DataArts Studio workspace accessible).
Procedure 2. Verify the masking rule for <code>customers.ssn</code> is active and uses the Mask All strategy. 3. Query the <code>customers</code> table as the test user via DataArts Studio SQL console. 4. Verify the <code>ssn</code> column returns masked values (e.g. <code>*****</code>) instead of real data. 5. Query the same table as an admin user and verify real values are returned.	1. In the DataArts Studio console, navigate to Data Architecture → Data Masking.
Expected Result 2. Non-admin	1. Masking rule is active for <code>customers.ssn</code>.



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Environment Setup and Tooling



5.1 TC-009: Verify Terraform infrastructure deployment

Objective	Confirm that the Terraform configuration deploys all required infrastructure resources (VPC, MRS, OBS, DataArts Studio) without errors and the state file reflects the expected resources.
Prerequisites	1. terraform CLI installed (version $\geq$ 1.0). 2. AK/SK credentials configured via environment variables or provider.tf. 3. Terraform configuration files in the project directory.
Procedure	1. Navigate to the Terraform project directory. 2. Run <code>terraform init</code> to initialize the working directory. 3. Run <code>terraform plan</code> and review the planned changes — confirm all expected resources are listed. 4. Run <code>terraform apply -auto-approve</code> and wait for completion. 5. Run <code>terraform output</code> and verify all output values are populated. 6. Run <code>terraform state list</code> and confirm the



5.2 TC-010: Verify test data upload to OBS

Objective	Confirm that all required test data files (CSV, JSON) can be uploaded to the designated OBS bucket and are accessible by the DataArts Studio pipeline.
Prerequisites	1. TC-002 completed (OBS bucket accessible).
2. Test data files prepared in the local directory <code>./test-data/</code>. 3. <code>obsutil</code> CLI configured with the test user credentials.	
Procedure	1. List the local test data files and verify their integrity (row count, schema).
2. Upload all files to the OBS raw bucket using <code>obsutil</code>.	
3. Verify each file exists in OBS by listing the bucket contents.	
4. Download a sample file from OBS and compare it with the local original (checksum).	
5. In DataArts Studio, verify the data source connection can read the uploaded files.	
<pre># Upload test data files to OBS obsutil cp ./test-da</pre>	1. All test data files are uploaded without errors.

[Image placeholder]

File: assets/test-execution-dashboard.png

Description: Screenshot of the test execution dashboard showing all test case results

Place the image file at the path above, then replace this command with

*`\image{assets/test-execution-dashboard.png}` or
`\imagecap{assets/test-execution-dashboard.png}{caption}`.*

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Changelog

1.1.0

September 12, 2026

- Added comprehensive feature demonstrations: `warning`, `tip`, `infobox` callouts; code blocks; `badge`; `weblink`; `note`; `param`; `image`, `imagecap`, and `imageplaceholder`; menu paths; and `inlinecode` references.
- Added Security and Access Control test cases (TC-007, TC-008).
- Added Environment Setup and Tooling test cases (TC-009, TC-010).

1.0.0

September 12, 2026

- Initial version.
 - Added Platform Architecture test cases (TC-001 to TC-003).
 - Added Data Engineering test cases (TC-004 to TC-006).
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